LTE Physical Activity Analysis

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Study Summary

Klinefelter syndrome (KS), 47,XXY, affects 1 in 600 males and is characterized by an additional X chromosome. Men with KS have altered testicular development, deficient testosterone levels, and infertility. They are less physically active than other men, have higher BMIs on average, and report higher levels of fatigue. However, little research has explored the relationship between sleep and activity in KS. This analysis analyzes differences in activity and sleep outcomes between KS cases and non-KS age- and BMI-matched controls.

KS males and non-KS controls aged 15 - 27 were recruited into the Lipids to Energy (LTE) clinical trial. Groups were age- and BMI- matched. Participants wore ActiGraph accelerometers and ActiWare sleep watches for one continuous week while completing daily surveys on activity, sleep, and diet. ActiGraph accelerometers monitor activity including measures of body position and movement. ActiWare sleep watches monitor sleep and wake time as well as light exposure.
Our analysis includes 19 KS and 18 controls with sufficient activity and sleep watch data over a 7-day time span.

Data Processing

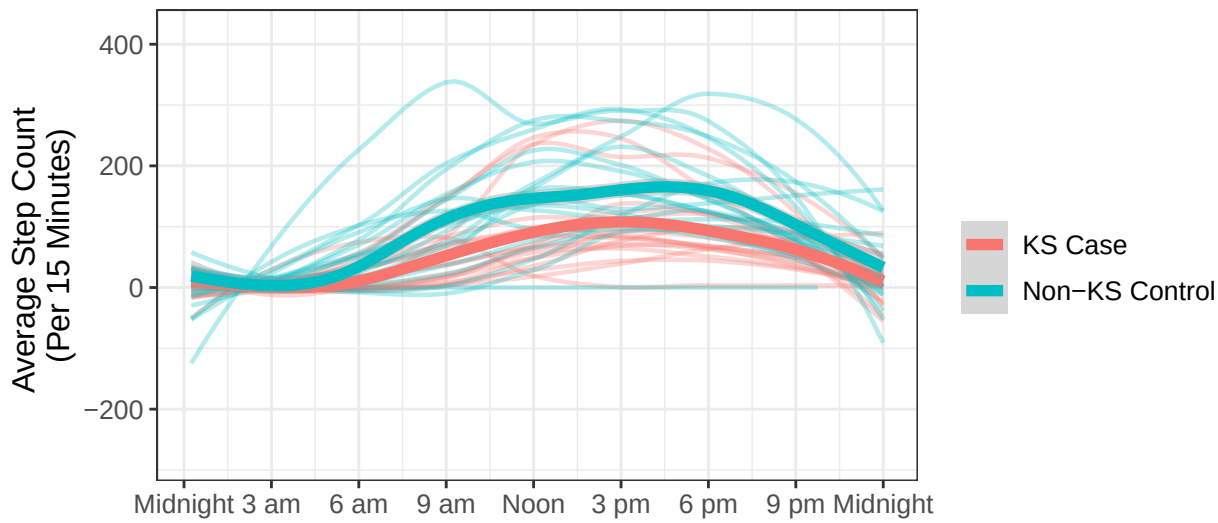
The first and last day of wear for each participant were excluded for analysis.

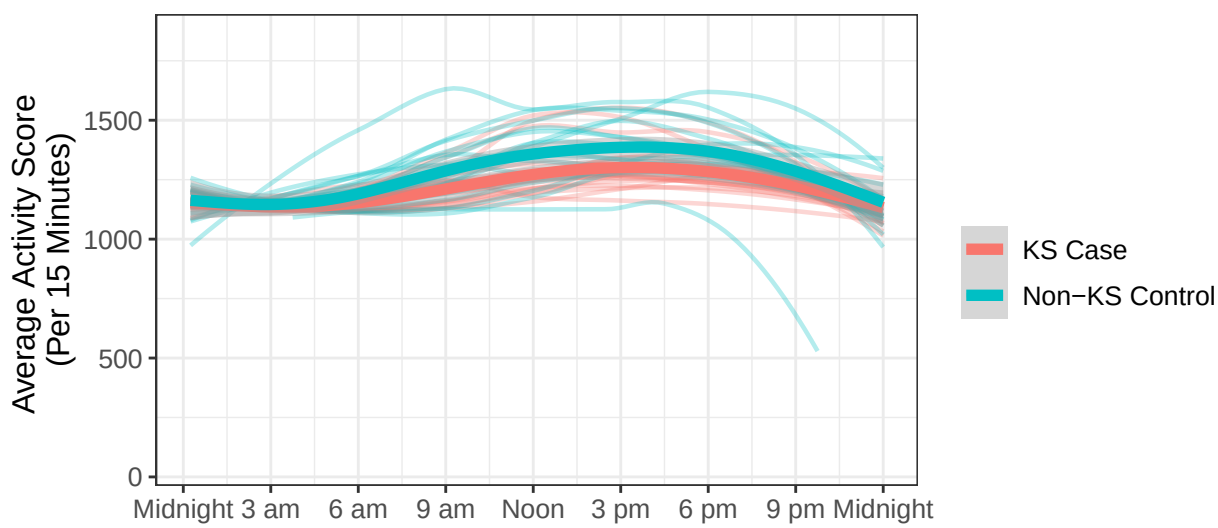
Data were extracted from ActivPal in **15 second epochs** but summarized within minutes.

Entries where the watch was indicated to not have been worn were removed. Only days with >70% of wear time were included.

Preliminary Plots

Aggregated Across Days





Statistical Methods

We model minute-level step count data using a functional regression framework (function-on-scalar regression) with subject-specific random functional intercepts.

Let $Y_i(t)$ denote the step count for subject $i = 1, ..., N$ at time t (15 minute interval of the day). The mean structure was modeled with smooth functions of time and group differences:

$$Y_i(t) = \beta_0(t) + \beta_{group}(t)\mathbf{1}_{group=KS\ Case} + b_i(t) + \epsilon_i(t)$$

,

where

$\beta_0(t)$ is the smoothed intercept

$\beta_{group}(t)$ is the smoothed cofficient for the difference between groups

$\beta_i(t)$ is the subject-specific random functional intercept

$\epsilon_i(t)$ is the residual error

Questions for Methods

- The standard data preprocessing step has been to average measures at the same minute across days but I believe we could lose a lot of information this way. Are there FDA methods using linear mixed effects model methodology to account for repeated days within an individual?
  - Random effect for subject with a fixed effect for day. Covariance structure would likely be unstructured for days within subject.
  - If we were able to use LMM methods, do we anticipate a Hawthorne effect may exist for people who start using the watches? In this analysis, the first day of data were removed but primarily because they were incomplete days.

Results

```r
#### FDA
## Make Wide datasets
long_15min <- sumdata_15min[,c("ID", "Avg_StepCount", "index")] %>%
  pivot_wider(names_from = index, values_from = Avg_StepCount)
```

Simple Linear Fit

```r
tbl <- tbl_regression(lmer(Avg_StepCount ~ group + (1|ID), data = sumdata_15min)) |>
  as_gt() |>
  gt::tab_options(table.layout = "fixed")

knitr::asis_output(as.character(gt::as_latex(tbl)))
```

Characteristic	Beta	95% CI
group		
KS Case	—	—
Non-KS Control	39	13, 65

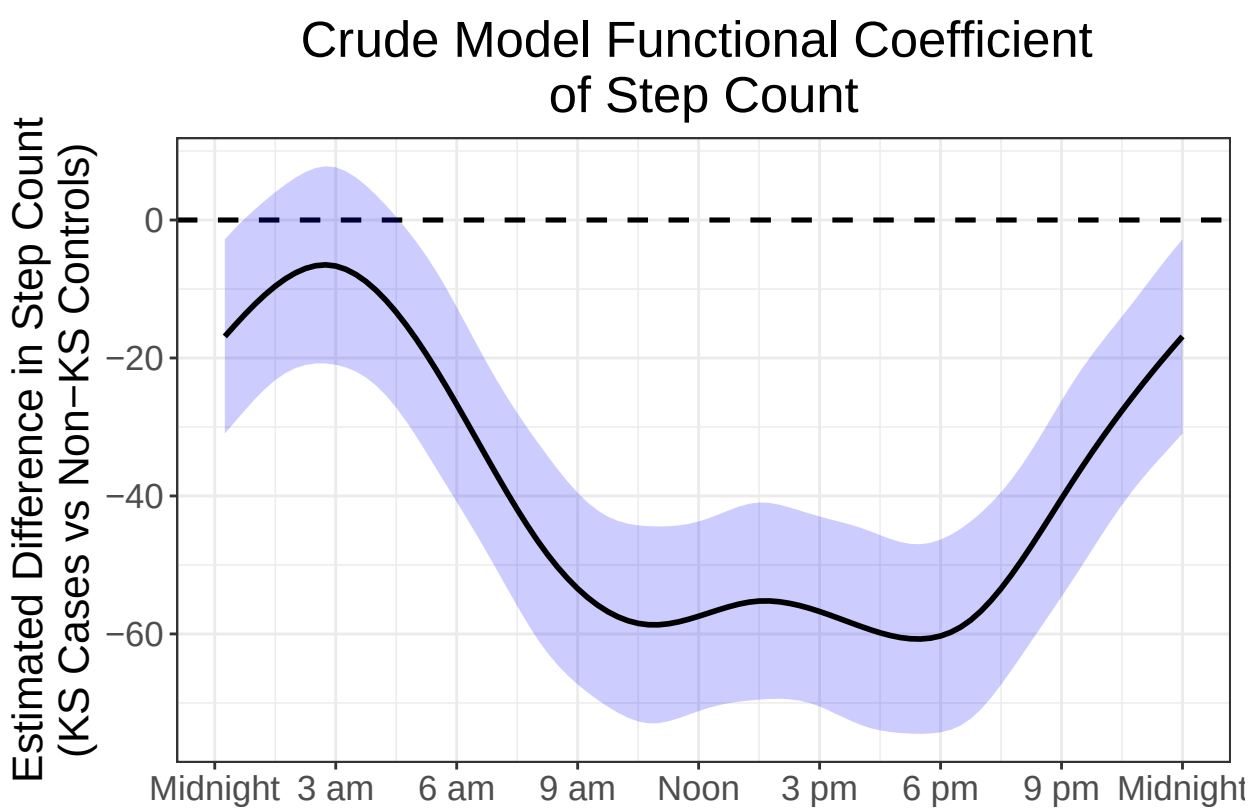
Abbreviation: CI = Confidence Interval

Functional Fit

```
##### fit the model for average step count across days within participant
sumdata_15min$group <- as.numeric(sumdata_15min$group == "KS Case")
sumdata_15min$Avg_StepCount_log <- log1p(sumdata_15min$Avg_StepCount)

# fit_15min <- gam(list(Avg_StepCount ~ s(index, bs="cc") + s(index, by = group, bs="cc"), ~ 1,
#                   family = ziP(),
#                   data=sumdata_15min)

fit_15min <- gam(
  Avg_StepCount ~ s(index, bs = "cc") + s(index, by = group, bs = "cc"),
  #family = tw(), # estimates the ML inside GAM
  data = sumdata_15min,
  method = "REML"
)
# tbl <- tbl_regression(fit_1min)

# Save gam residuals
resid_dat <- data.frame(cbind(".id" = sumdata_15min$ID,
                              ".index" = sumdata_15min$index,
                              ".value" = fit_15min$residuals))

## Convert all columns to numeric
resid_dat <- sapply( resid_dat, as.numeric )
resid_dat <- data.frame(resid_dat)


resid_dat_wide <- resid_dat %>%
  group_by(.id, .index) %>%
  slice(1) %>%
  ungroup() %>%
  pivot_wider(id_cols = .id, names_from = .index, values_from = .value) %>%
  dplyr::arrange(.id)

# Extract eigenfunctions
# step_mat <- as.matrix(step_wide[,-c(1:4)])
#resid_fpca <- fpca.sc(ydata = resid_dat)
```

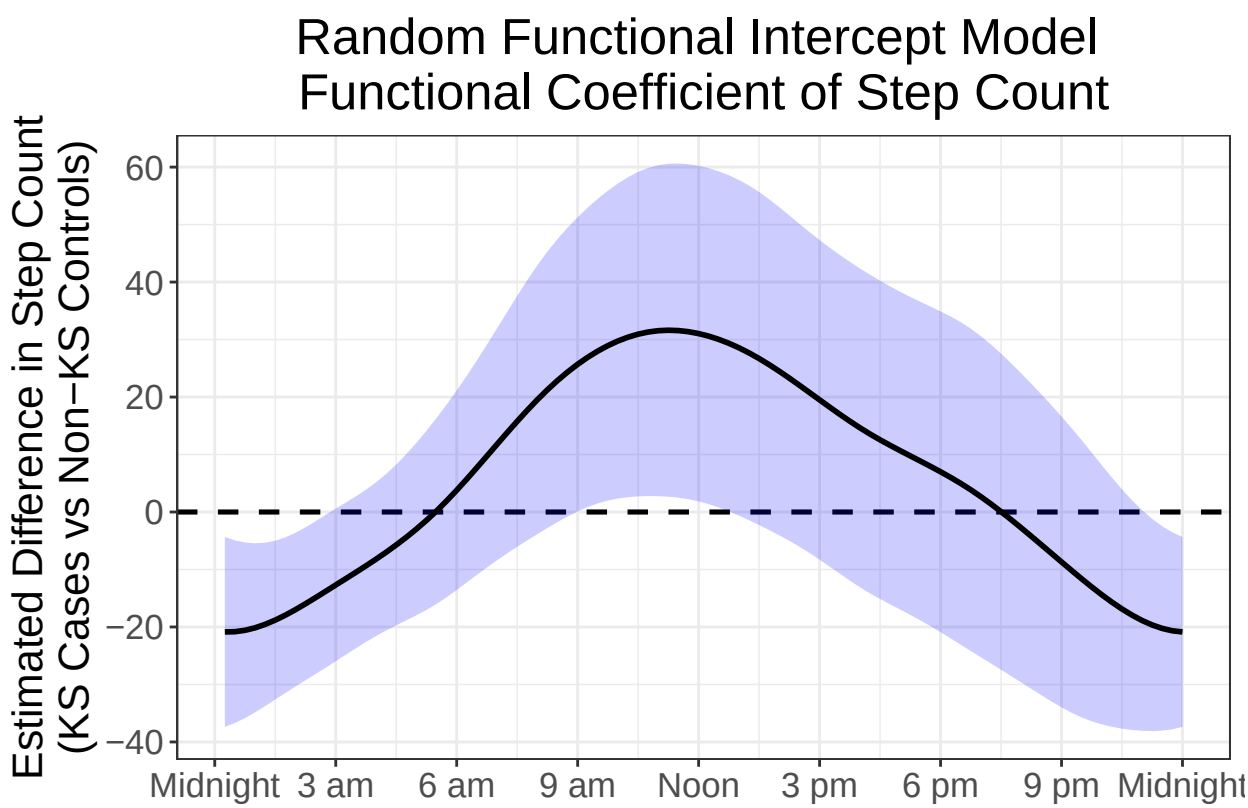
```r
resid_fpca <- fpca.face(as.matrix(resid_dat_wide))
Phi_hat <- resid_fpca$efunctions


#refund.shiny::plot_shiny(resid_fpca) # Plot to see eigenfunctions

# Include first 4 eigenfunctions in data
unique_index <- sort(unique(sumdata_15min$index))
# Check that Phi_hat has the same number of rows as unique_sind
#stopifnot(nrow(Phi_hat) == length(unique_index))
# Match each observation's sind to the corresponding eigenfunction value
sumdata_15min$Phi1 <- Phi_hat[match(sumdata_15min$index, unique_index), 1]*-1
sumdata_15min$Phi2 <- Phi_hat[match(sumdata_15min$index, unique_index), 2]
sumdata_15min$Phi3 <- Phi_hat[match(sumdata_15min$index, unique_index), 3]
sumdata_15min$Phi4 <- Phi_hat[match(sumdata_15min$index, unique_index), 4]

sumdata_15min$ID <- as.numeric(sumdata_15min$ID )

# Random functional intercept model
fit_15min_rfi <- gam(Avg_StepCount ~ s(index, bs="cc") +
                  s(index, by = group, bs="cc") +
                  s(ID, by = Phi1, bs="cc") +
                  s(ID, by = Phi2, bs="cc") +
                  s(ID, by = Phi3, bs="cc") +
                  s(ID, by = Phi4, bs="cc"),
                  #family = tw(),
                  data=sumdata_15min,
                  method = "REML")

rfi_summary <- data.frame(summary(fit_15min_rfi)$s.table)

## Prediction data
df_pred <- data.frame(index = seq(1, 96, by = 1),
                      ID = sumdata_15min$ID[1],
                      group = 1, Phi1 = 0, Phi2 = 0, Phi3 = 0, Phi4 = 0)
fhat_naive <- predict(fit_15min, newdata=df_pred, se.fit=TRUE,type='terms')
fhat_rfi   <- predict(fit_15min_rfi, newdata=df_pred, se.fit=TRUE,type='terms')

## Save crude and rfi estimates
group_hat <- fhat_naive$fit[,2]; group_se <- fhat_naive$se.fit[,2]
group_hat_rfi <- fhat_rfi$fit[,2]; group_se_rfi <- fhat_rfi$se.fit[,2]
crude_ests <- data.frame(group_hat,
                        group_low = group_hat - 1.96*group_se,
                        group_high = group_hat + 1.96*group_se,
                        sind = seq(1, 96, by = 1))
rfi_ests <- data.frame(group_hat_rfi,
                        group_low = group_hat_rfi - 1.96*group_se_rfi,
```

```
                        group_high = group_hat_rfi + 1.96*group_se_rfi,
                        sind = seq(1, 96, by = 1))

# Make plots
ggplot(crude_ests, aes(x = sind, y = group_hat)) +
  geom_hline(yintercept = 0, color = "black", linetype = "dashed", size = 0.9) +
  geom_ribbon(aes(x = sind, ymin = group_low, ymax = group_high),
              fill = "blue", alpha = 0.2) +
  geom_line(size = 1)  +
  ylab("Estimated Difference in Step Count\n(KS Cases vs Non-KS Controls)") +
  ggtitle("Crude Model Functional Coefficient \nof Step Count") + theme_bw() +
  scale_x_continuous(breaks = c(0, 12, 24, 36, 48, 60, 72, 84, 96),
                     labels = c("Midnight", "3 am", "6 am", "9 am", "Noon", "3 pm", "6 pm",
                                "9 pm", "Midnight")) +
  xlab("") + theme(text = element_text(size = 16), plot.title = element_text(hjust = 0.5))
```




```
ggplot(rfi_ests, aes(x = sind, y = group_hat_rfi)) +
  geom_hline(yintercept = 0, color = "black", linetype = "dashed", size = 0.9) +
  geom_ribbon(aes(x = sind, ymin = group_low, ymax = group_high),
              fill = "blue", alpha = 0.2) +
  geom_line(size = 1)  +
  ylab("Estimated Difference in Step Count\n(KS Cases vs Non-KS Controls)") +
```

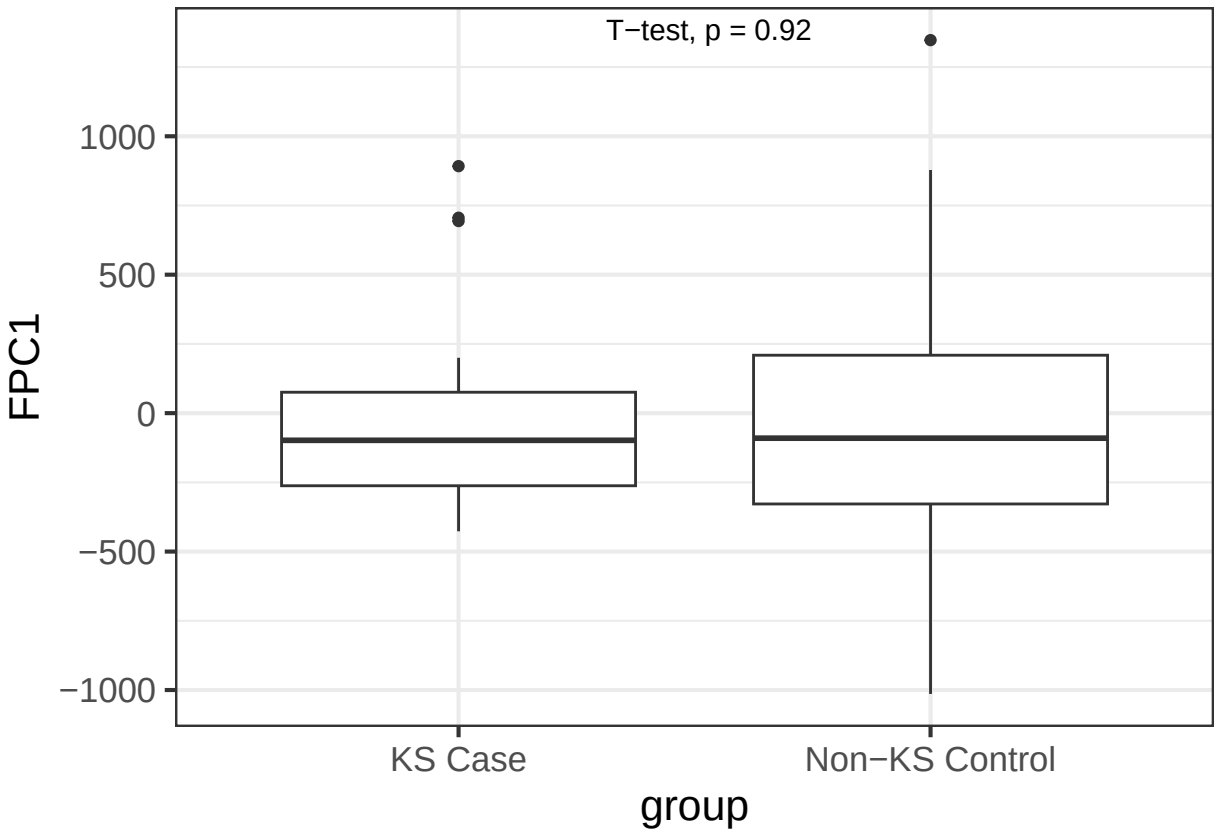
```
  ggtitle("Random Functional Intercept Model \nFunctional Coefficient of Step Count") +
  theme_bw() +
  scale_x_continuous(breaks = c(0, 12, 24, 36, 48, 60, 72, 84, 96),
                     labels = c("Midnight", "3 am", "6 am", "9 am", "Noon", "3 pm", "6 pm",
                                "9 pm", "Midnight")) +
  xlab("") + theme(text = element_text(size = 16), plot.title = element_text(hjust = 0.5))
```



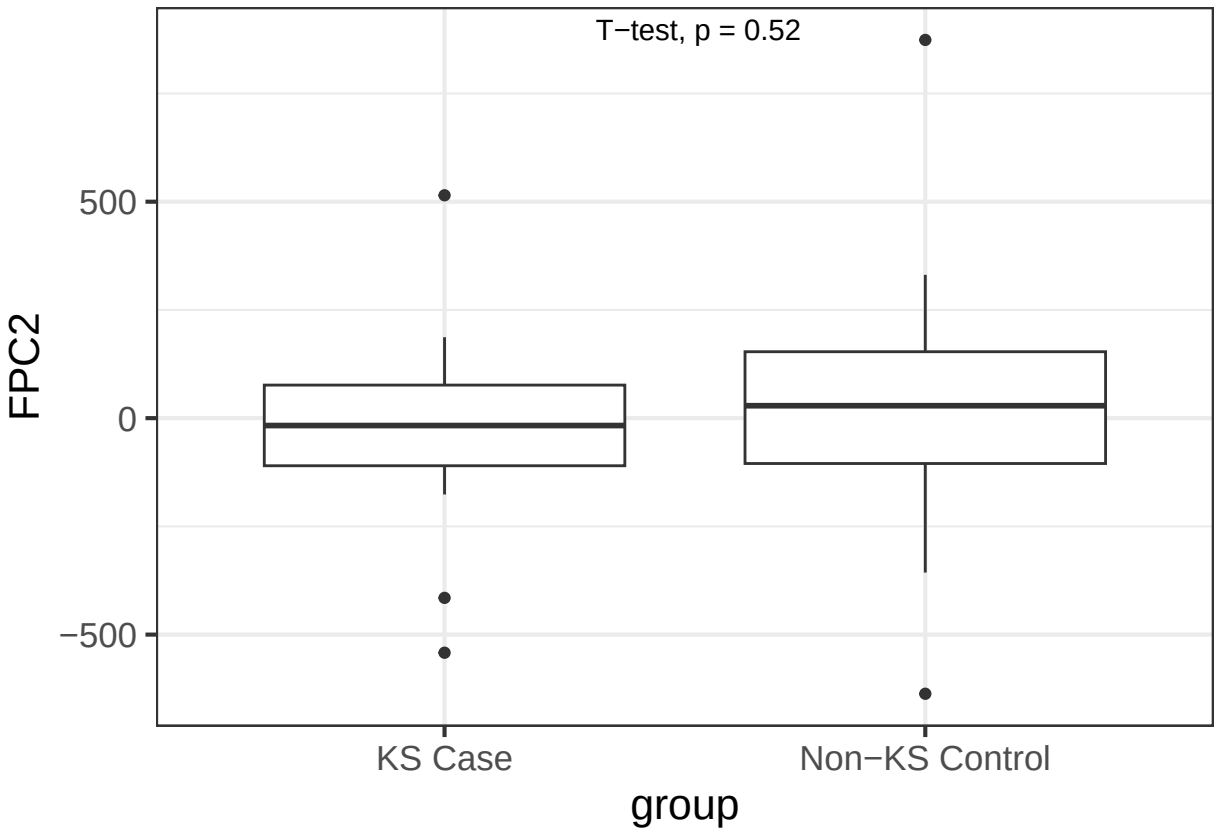
Correlation between scores and group

```
# Extract the scores from fpca.face
scores <- data.frame(resid_fpca$scores)
colnames(scores) <- paste0("FPC", 1:ncol(scores))
scores$ID <- resid_dat_wide$.id

# Add group info
scores$group <- sumdata_1min$group[match(scores$ID, sumdata_1min$ID)]
#scores$group <- ifelse(scores$group == 1, "KS Case", "Non-KS Control")

scores %>%
  mutate(group = as.factor(group)) %>%
```

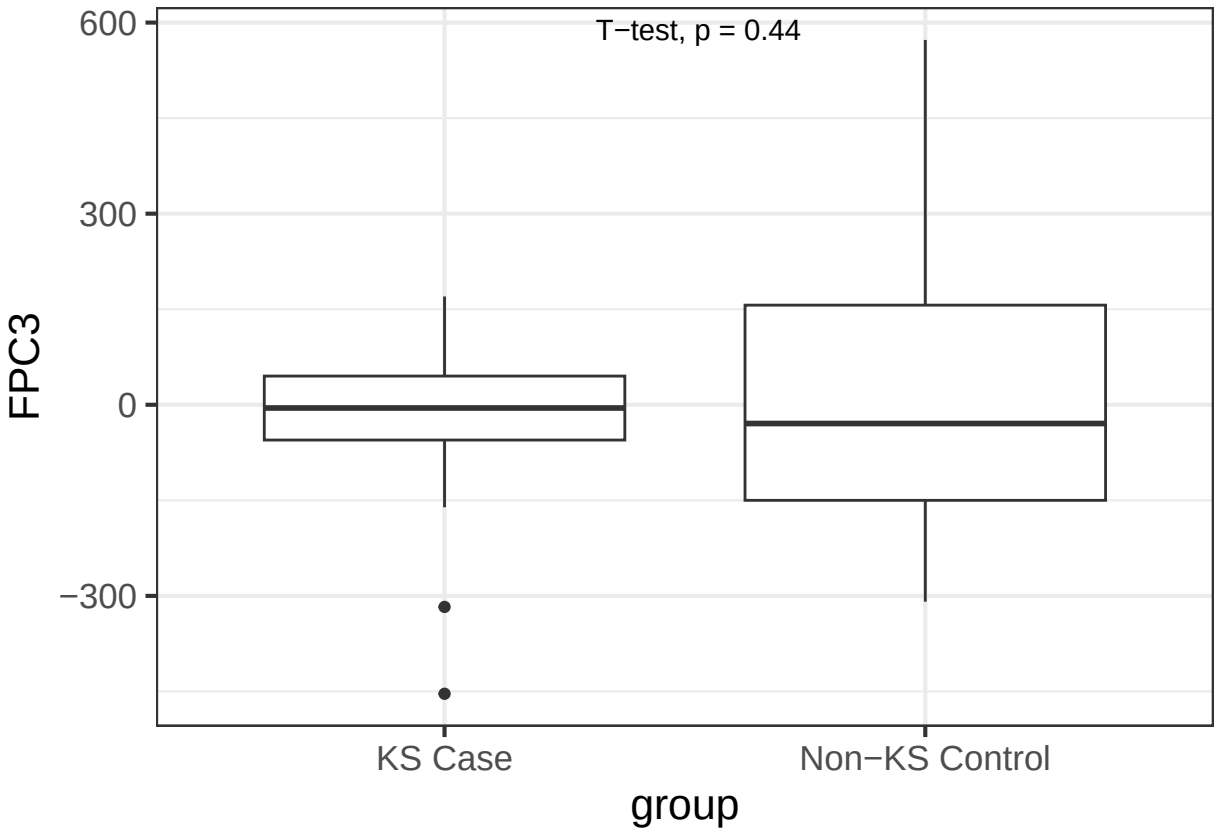
```
  ggplot(aes(x = group, y = FPC1)) +
  geom_boxplot() +
  stat_compare_means(method = "t.test", label.x = 1.4) +
  theme_bw(base_size = 16)
```



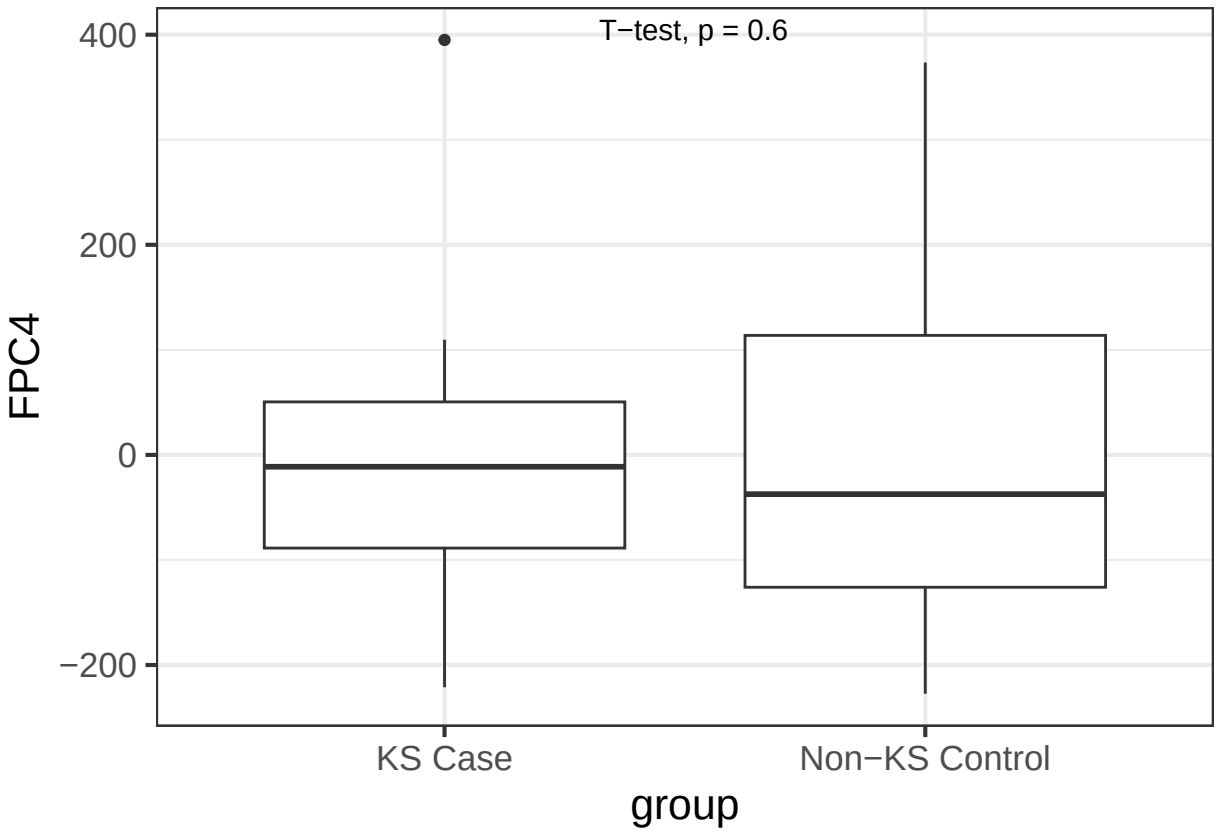

```
scores %>%
  mutate(group = as.factor(group)) %>%
  ggplot(aes(x = group, y = FPC2)) +
  geom_boxplot() +
  stat_compare_means(method = "t.test", label.x = 1.4) +
  theme_bw(base_size = 16)
```




```
scores %>%
  mutate(group = as.factor(group)) %>%
  ggplot(aes(x = group, y = FPC3)) +
  geom_boxplot() +
  stat_compare_means(method = "t.test", label.x = 1.4) +
  theme_bw(base_size = 16)
```




```
scores %>%
  mutate(group = as.factor(group)) %>%
  ggplot(aes(x = group, y = FPC4)) +
  geom_boxplot() +
  stat_compare_means(method = "t.test", label.x = 1.4) +
  theme_bw(base_size = 16)
```


T−test, p = 0.6
400
200
0
−200
FPC4
KS Case
Non−KS Control
group